

Why do we regularise in every iteration?

Evangelos Papoutsellis¹, Zeljko Kereta², Kostas Papafitsoros³

¹Finden Ltd, ²University College London, ³Queen Mary University of London

Iterative Regularisation Methods

- The majority of the algorithms for imaging inverse problems involve the evaluation of the proximal operator in every iteration.
- Proximal operator can be related to the regularisation term.
- Significant computational overhead since such evaluation can be costly, e.g., TV, TGV, TNV and many more.
- Proximal Gradient and Primal-Dual methods can be improved significantly in terms of CPU time.

Algorithm 1 GD

```
1: Parameters:  $\gamma > 0$ 
2: Initialize:  $\mathbf{x}_0 \in \mathbb{X}$ 
3: for  $k = 0, \dots, K-1$  do
4:    $\mathbf{x}_{k+1} = \mathbf{x}_k - \gamma \nabla f(\mathbf{x}_k)$ 
5: end for
```

Algorithm 2 PGD/ISTA/FBS

```
1: Parameters:  $\gamma > 0$ 
2: Initialize:  $\mathbf{x}_0 \in \mathbb{X}$ 
3: for  $k = 0, \dots, K-1$  do
4:    $\mathbf{x}_{k+1} = \text{prox}_{\gamma g}(\mathbf{x}_k - \gamma \nabla f(\mathbf{x}_k))$ 
5: end for
```

Algorithm 3 FISTA

```
1: Parameters:  $\gamma > 0, t_0 = 1$ 
2: Initialize:  $\mathbf{x}_0 \in \mathbb{X}$ 
3: for  $k = 0, \dots, K-1$  do
4:    $t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}, a_k = \frac{t_{k-1}}{t_{k+1}}$ 
5:    $\bar{\mathbf{x}}_{k+1} = \mathbf{x}_k + a_k(\mathbf{x}_k - \mathbf{x}_{k+1})$ 
6:    $\mathbf{x}_{k+1} = \text{prox}_{\gamma g}(\bar{\mathbf{x}}_{k+1} - \gamma \nabla f(\bar{\mathbf{x}}_{k+1}))$ 
7: end for
```

Acquired Data

$$\mathbf{b} = \mathbf{A}\mathbf{u}^\dagger + \boldsymbol{\eta}$$

Total Variation Regularisation

$$\min_{\mathbf{u}} \left\{ \alpha \text{TV}(\mathbf{u}) + \frac{1}{2} \|\mathbf{A}\mathbf{u} - \mathbf{b}\|_2^2 \right\}$$

Optimisation Frameworks

$$\min_{\mathbf{x} \in \mathbb{X}} f(\mathbf{x}) + g(\mathbf{x})$$

$$\min_{\mathbf{x} \in \mathbb{X}} f(\mathcal{K}\mathbf{x}) + g(\mathbf{x})$$

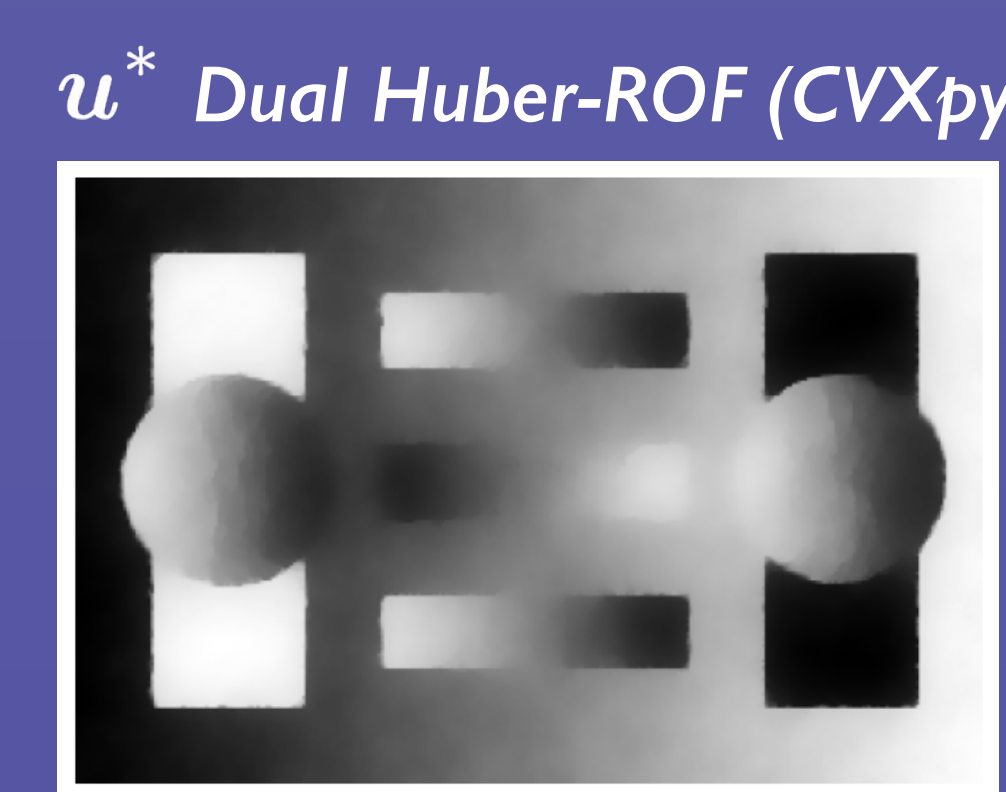
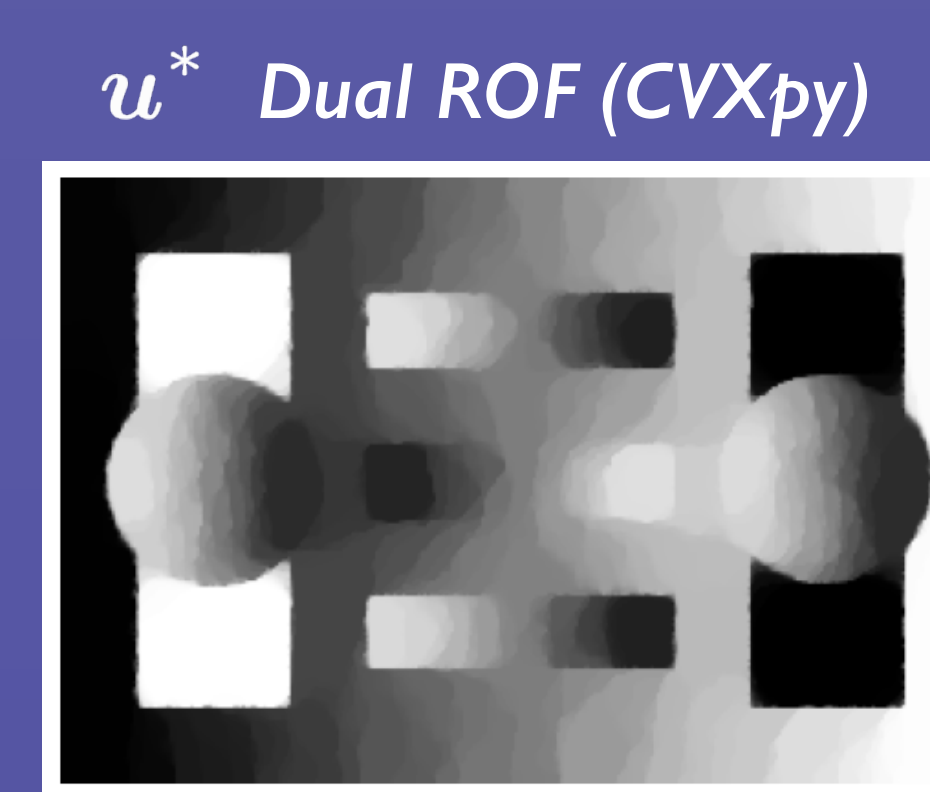
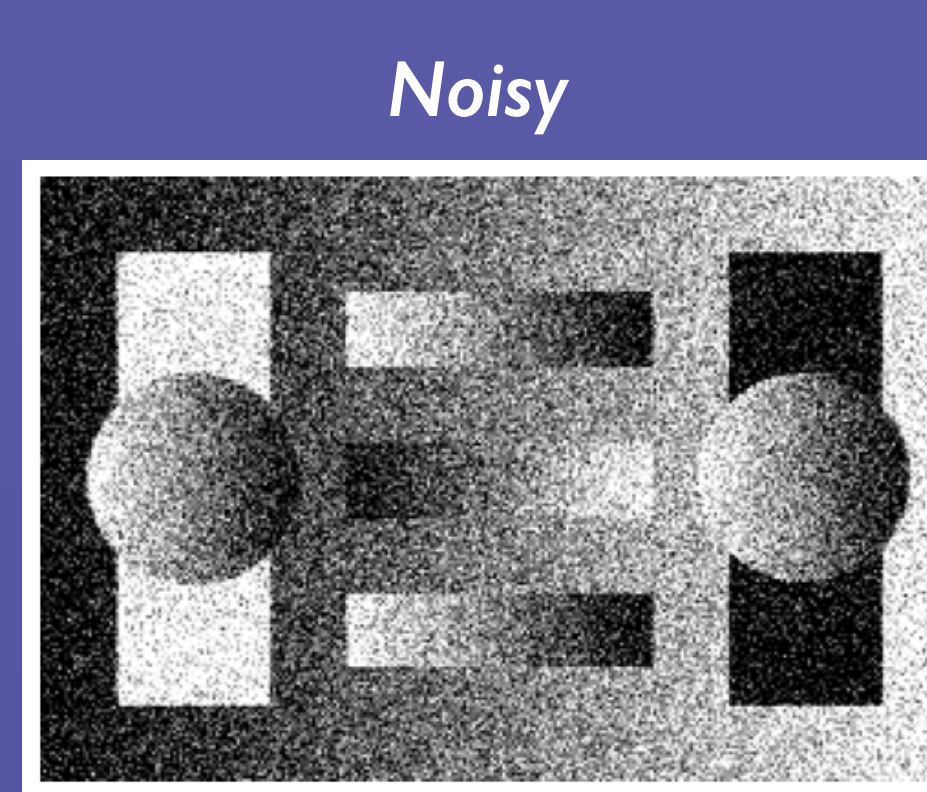
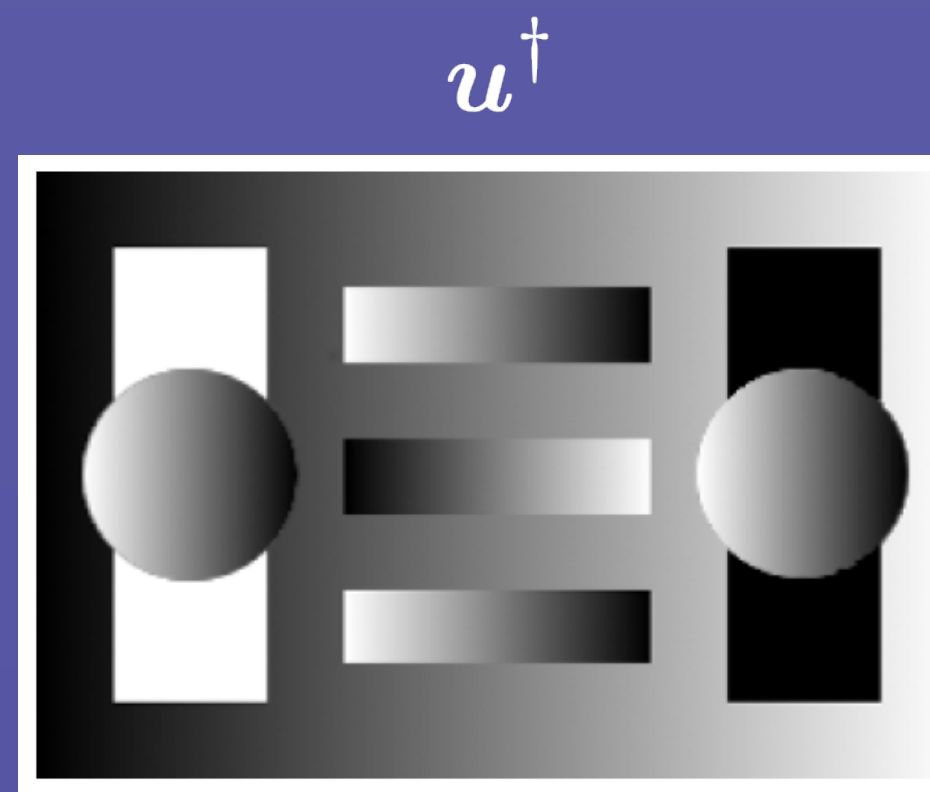
But Why?

- Actually, no need to regularise in every iteration.
- Proximal operator is computed based on probability p .
- Computational gain without affecting the convergence.

Algorithm 4 ProxSkip

```
1: Parameters:  $\gamma > 0$ , probability  $p > 0$ 
2: Initialize:  $\mathbf{x}_0, \mathbf{h}_0 \in \mathbb{X}$ 
3: for  $k = 0, 1, \dots, K-1$  do
4:    $\hat{\mathbf{x}}_{k+1} = \mathbf{x}_k - \gamma(\nabla f(\mathbf{x}_k) - \mathbf{h}_k)$ 
5:   if  $\text{Prob}(\theta_k = 1) = p$  then
6:      $\mathbf{x}_{k+1} = \text{prox}_{\frac{\gamma}{p}g}(\hat{\mathbf{x}}_{k+1} - \frac{\gamma}{p}\mathbf{h}_k)$ 
7:   else  $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1}$ 
8:   end if
9:    $\mathbf{h}_{k+1} = \mathbf{h}_k + \frac{p}{\gamma}(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1})$ 
10: end for
```

Total Variation Denoising



ROF model

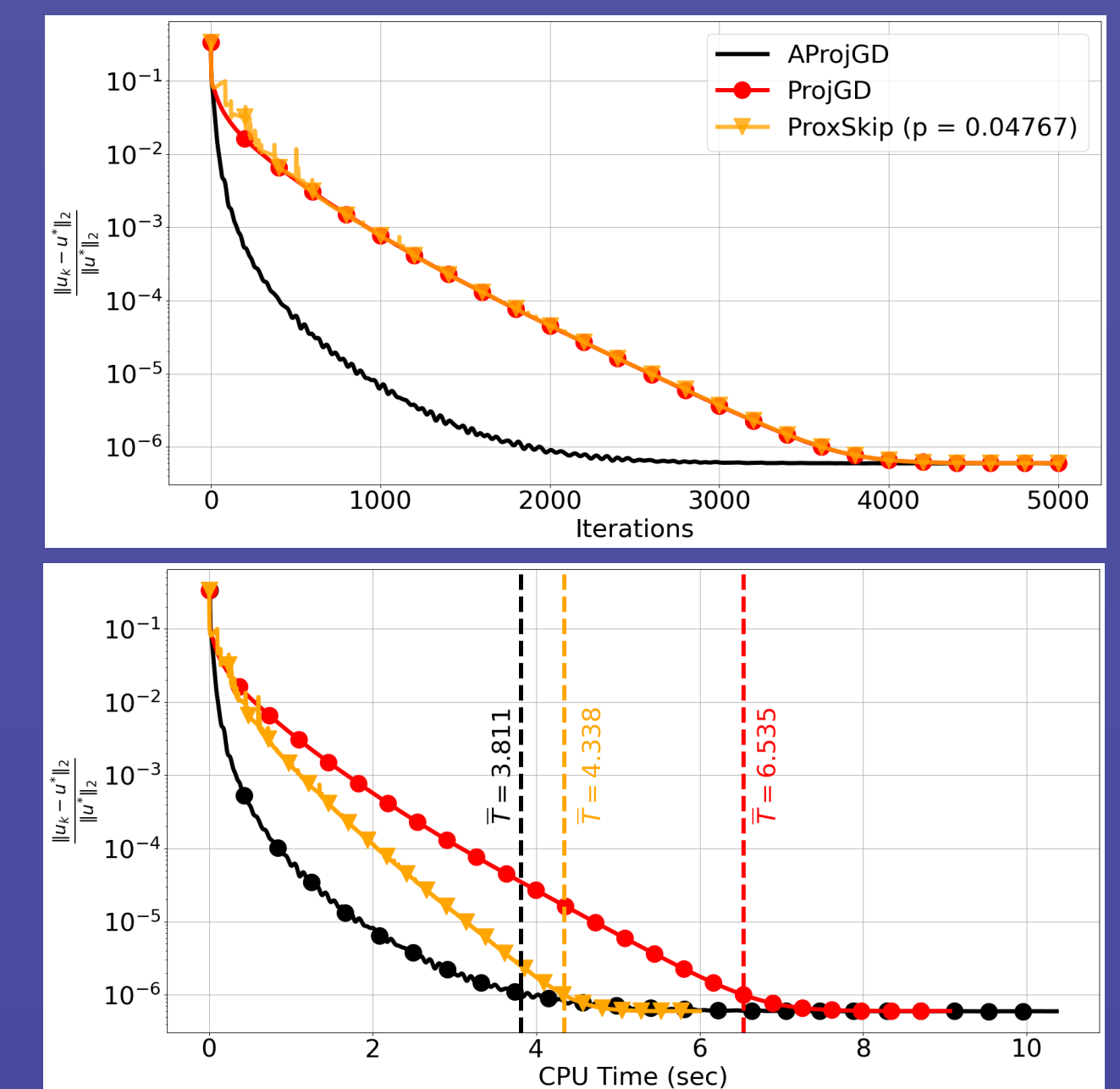
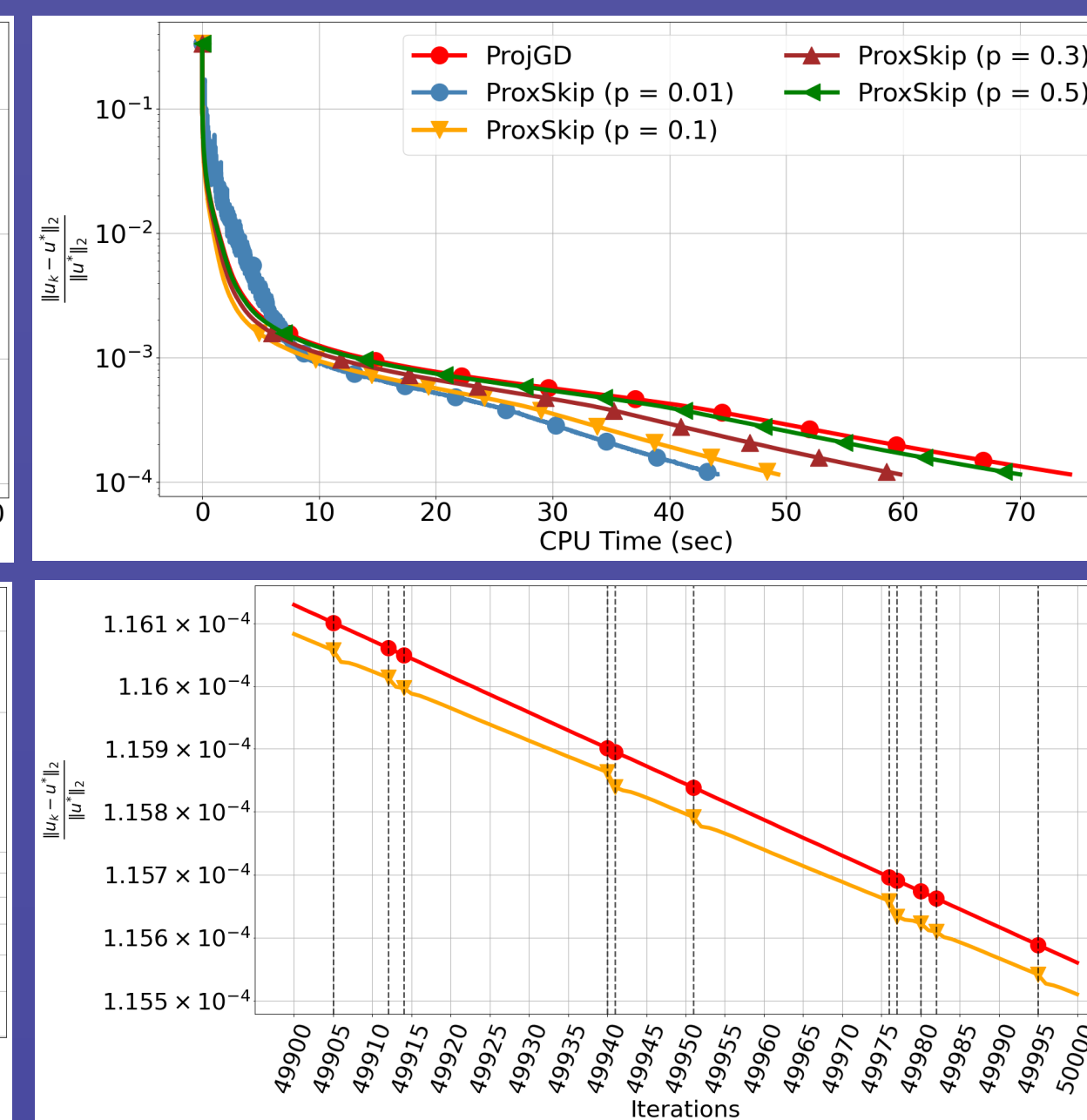
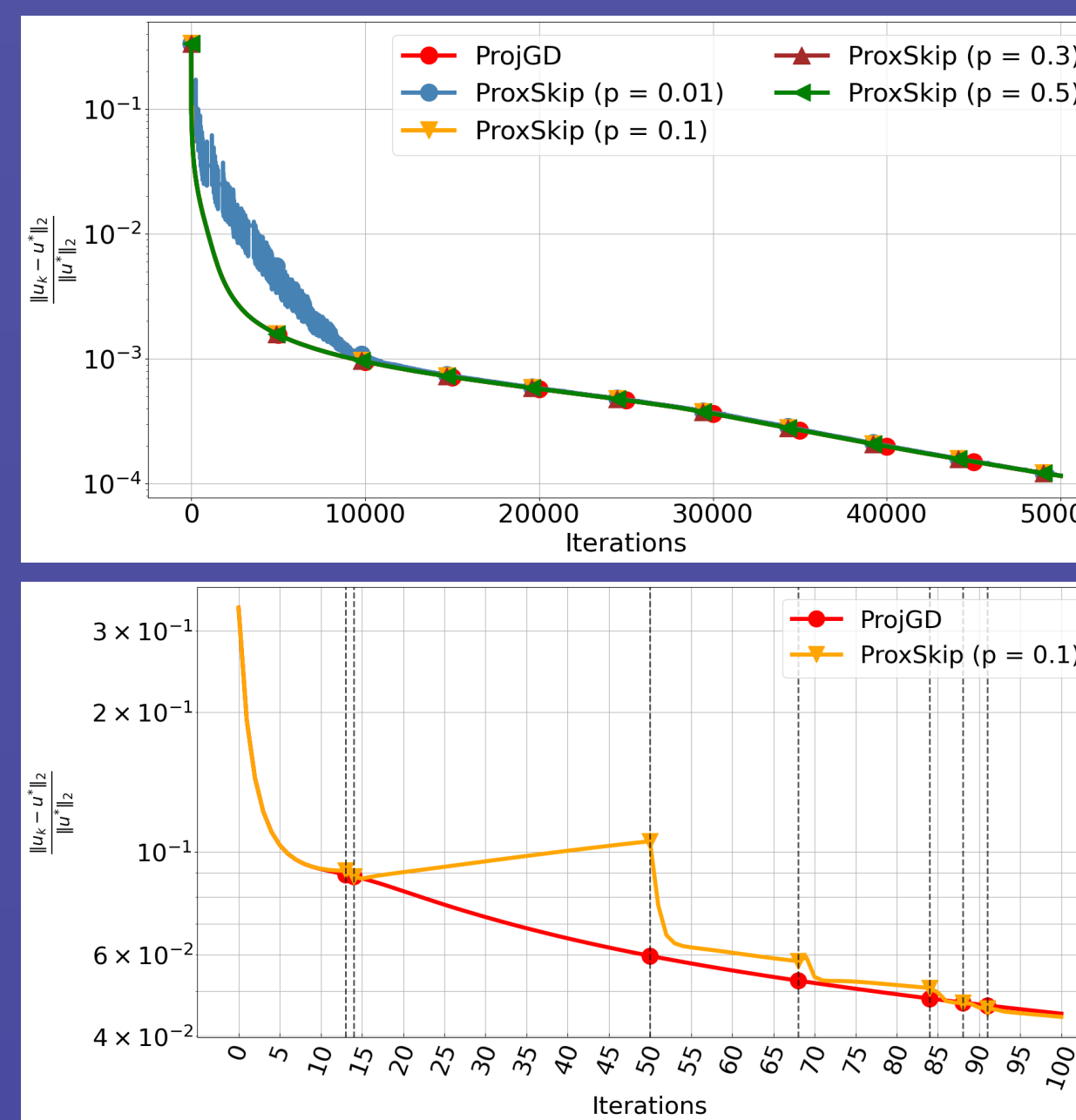
$$\min_{\mathbf{u}} \left\{ \alpha \text{TV}(\mathbf{u}) + \frac{1}{2} \|\mathbf{u} - \mathbf{b}\|_2^2 \right\}$$

Dual ROF model

$$\min_{\|\mathbf{q}\|_{2,\infty} \leq \alpha} \left\{ \frac{1}{2} \|\text{div} \mathbf{q} + \mathbf{b}\|_2^2 + \frac{1}{2} \|\mathbf{b}\|_2^2 \right\}$$

Dual Huber-ROF model

$$\min_{\|\mathbf{q}\|_{2,\infty} \leq \alpha} \left\{ \frac{1}{2} \|\text{div} \mathbf{q} + \mathbf{b}\|_2^2 + \frac{1}{2} \|\mathbf{b}\|_2^2 + \frac{\varepsilon}{2\alpha} \|\mathbf{q}\|_2^2 \right\}$$



How often can we skip?

- Strong convexity is required to compute the optimal probability p , e.g., Dual Huber-ROF.

$$p \geq \sqrt{\frac{\mu}{L}} \Rightarrow \mathcal{O}\left(\frac{1}{\sqrt{p}} \log\left(\frac{1}{\varepsilon}\right)\right) \quad \text{Number of Proximal Evaluations (In expectation)}$$

Inexact regularisation

- Proximal operator has no analytic solution.
- Iterative (inner) solver is used up to some accuracy.
- Stopping criteria: a) iterations, b) distance error.

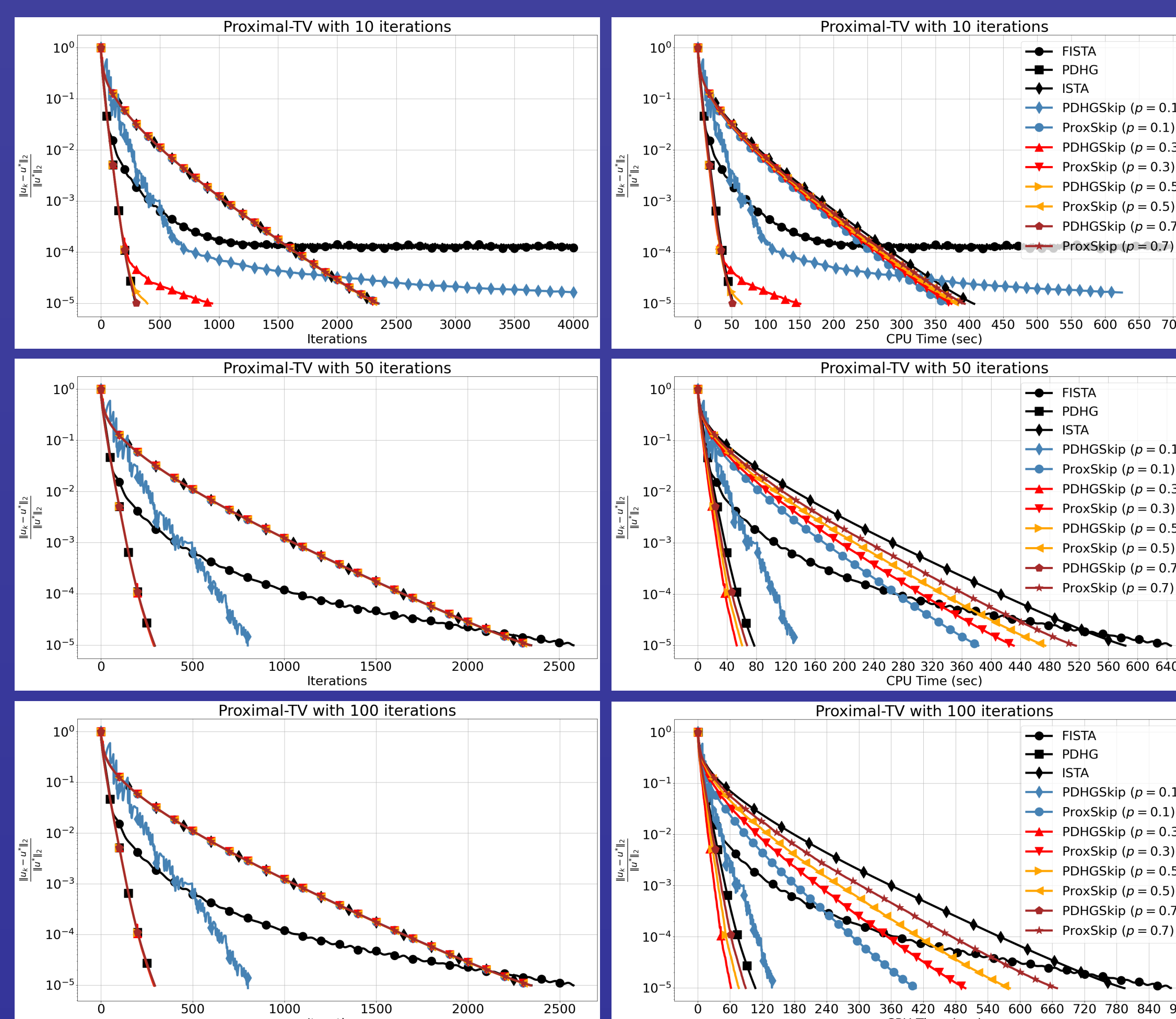
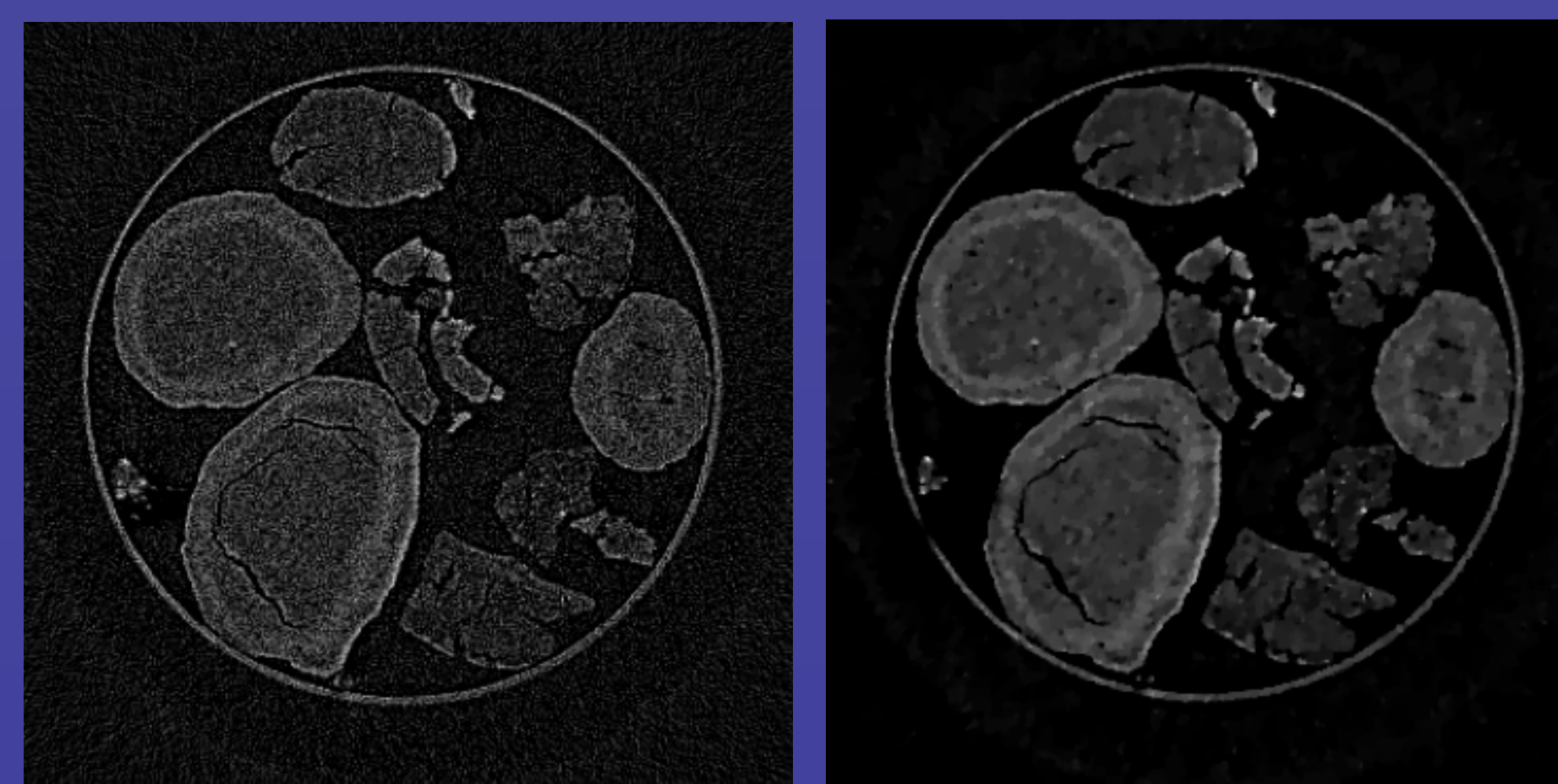
Open Source Libraries	Algorithm	Proximal TV	Stopping Criteria (Default)
Tomopy	PDHG	ROF	Iterations: User input
DeepInv	PDHG	ROF	Iterations = 1000 Distance Error
CIL	FISTA	Dual ROF	Iterations = 100
TIGRE	PDHG	Dual ROF	Iterations = 50
PyHST2	FISTA	Dual ROF	Iterations = 200 Primal dual gap

Primal-Dual Extension (PDHGSkip)

- ProxSkip can be extended to primal-dual algorithms.
- More flexible and does not require L-smoothness.
- Different formulations allow proximal skipping related to both regulariser and fidelity terms.

```
1: Parameters:  $\sigma, \tau > 0$ , probability  $p$ 
2: Initialize:  $\mathbf{x}_0, \mathbf{h}_0 \in \mathbb{X}, \mathbf{y}_0 \in \mathbb{Y}$ 
3: for  $k = 0, \dots, K-1$  do
4:    $\hat{\mathbf{x}}_k = \mathbf{x}_k - \sigma(\mathcal{K}^T \mathbf{y}_k - \mathbf{h}_k)$ 
5:   if  $\theta_k = 1$  then
6:      $\mathbf{x}_{k+1} = \text{prox}_{\frac{\sigma}{p}g}(\hat{\mathbf{x}}_k - \frac{\sigma}{p}\mathbf{h}_k)$ 
7:   else  $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_k$ 
8:   end if
9:    $\bar{\mathbf{x}}_{k+1} = 2\mathbf{x}_{k+1} - \mathbf{x}_k$ 
10:   $\mathbf{y}_{k+1} = \text{prox}_{\tau f^*}(\mathbf{y}_k + \tau \mathcal{K}\bar{\mathbf{x}}_{k+1})$ 
11:   $\mathbf{h}_{k+1} = \mathbf{h}_k + \frac{p}{\sigma}(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1})$ 
12: end for
```

Total Variation Tomography Reconstruction



Conclusion

Skipping proximal operators opens the door to revisiting a range of iterative regularisation methods for a plethora of imaging modalities, that were developed over the last decades, with the promising potential for significant CPU time reductions.

References

- Mishchenko, K., Malinovsky, G., Stich, S., Richtárik, P. ProxSkip: Yes! Local gradient steps provably lead to communication acceleration! Finally! <https://doi.org/10.48550/arXiv.2202.09357>
- Condat, L., Richtárik, P. Randprox: Primal-dual optimization algorithms with randomized proximal updates. ICLR, 2023. <https://doi.org/10.48550/arXiv.2207.12891>
- Papoutsellis, E., Kereta, Z., Papafitsoros, K. Why do we regularise in every iteration for imaging inverse problems? (preprint)
- Matras, D., Vamvakeros, A., et al. Multi-length scale 5D diffraction imaging of Ni-Pd/CeO₂-ZrO₂/Al₂O₃ catalyst during partial oxidation of methane. <https://doi.org/10.1039/D1TA01464A>
- Jorgensen, J.S., Ametova, E., et al. Core Imaging Library - part I: a versatile Python framework for tomographic imaging. <https://doi.org/10.1098/rsta.2020.0192>
- Papoutsellis, E., Ametova, E., et al. Core Imaging Library - part II: multichannel reconstruction for dynamic and spectral tomography. <https://doi.org/10.1098/rsta.2020.0193>
- Rasch, J., Chambolle, A. Inexact first-order primal-dual algorithms. <https://doi.org/10.1007/s10589-020-00186-y>

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